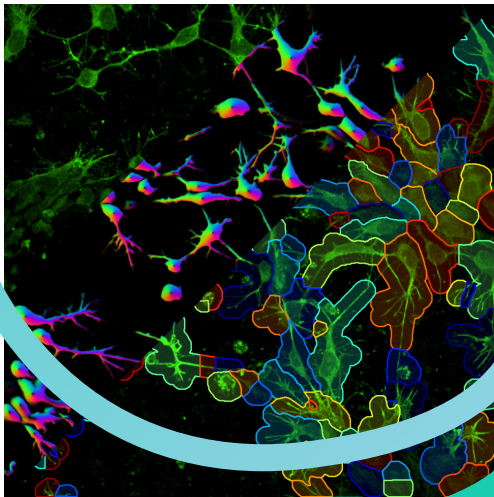


JUNE 2, 2026

Cellpose for machine learning in image analysis

Biological Analysis Unit – Institut Pasteur
Julie Mabon



summary

01. INSTANCE SEGMENTATION

02. CONVOLUTIONAL NEURAL NETWORKS

03. CONVOLUTIONAL NEURAL NETWORKS FOR INSTANCE
SEGMENTATION

04. PRACTICAL

PART 1

Instance Segmentation

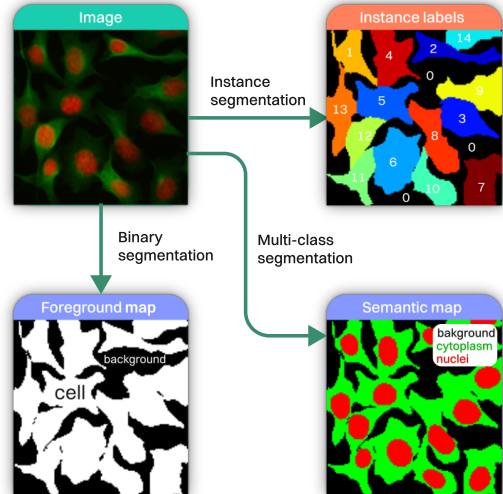
Instance Segmentation

Semantic Segmentation

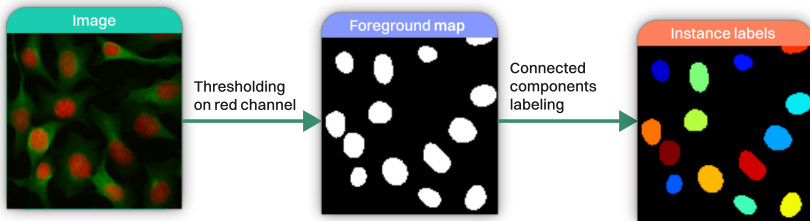
- Assigning a class/category to each pixel
- Binary : foreground/background
- Multiclass : cytoplasm/nuclei/bg

Instance Segmentation

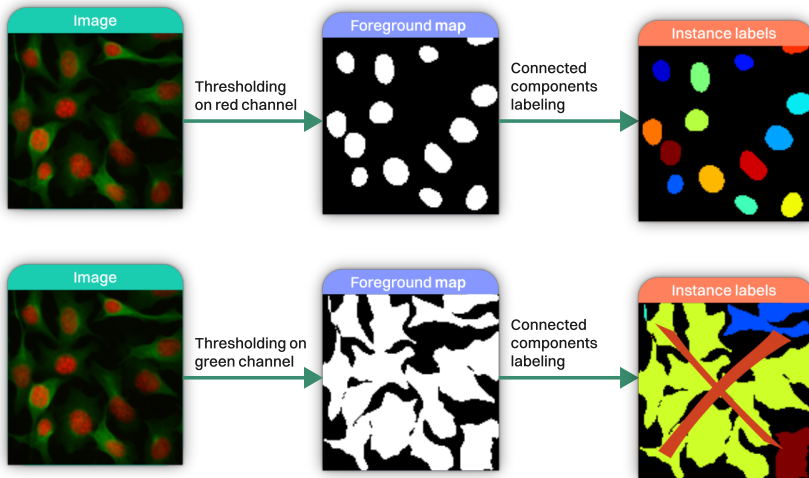
- Separating individual cells



The old reliable (?) method



The old reliable (?) method

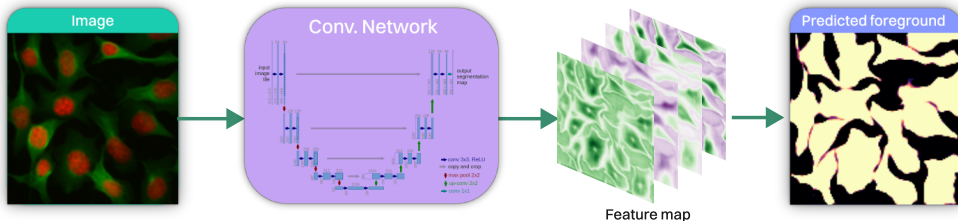


PART 2

Convolutional Neural Networks

Convolutional Neural Networks: Semantic segmentation

- **Convolutional Neural Networks** learn to extract **features** from images
- Relevant information is extracted from the **features**
- Image to image **translation** / **Local** by construction



Convolutional Neural Networks: Training

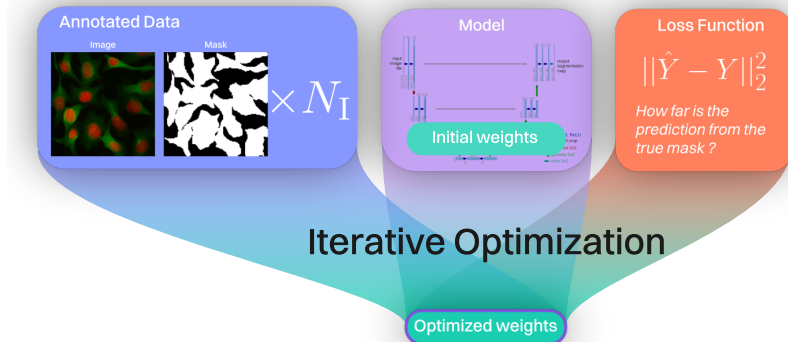
Training

Estimating the best weights given a bunch of **annotated data**, a **model** and a **loss function** (measure of the prediction error).

Convolutional Neural Networks: Training

Training

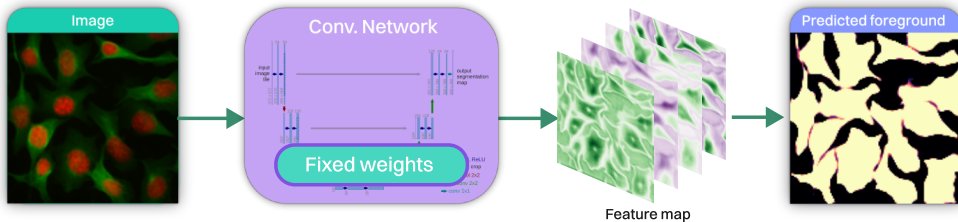
Estimating the best weights given a bunch of **annotated data**, a **model** and a **loss function** (measure of the prediction error).



Convolutional Neural Networks: Evaluation

Evaluation

Passing an image through the model with **fixed** weights/parameters.



Fine tuning

- A model is generally good on the same **domain** (eg similar looking images) that it was trained on
- We have a great **diversity** of imaging modalities / imaged samples
- You may need to **tune** the model to your specific task

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Fine-tuning

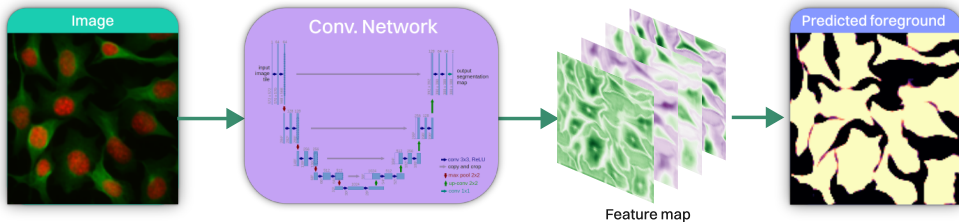
Further optimizing a **pre-trained** model on a **specific** task dataset.

PART 3

Convolutional Neural Networks for Instance segmentation

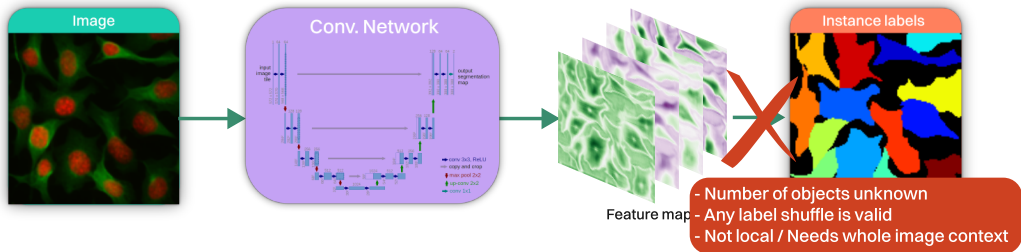
Convolutional Neural Networks: Instance segmentation

- Convolutional Neural Networks learn to extract features from images
- Relevant information is extracted from the features
- Image to image translation / Local by construction

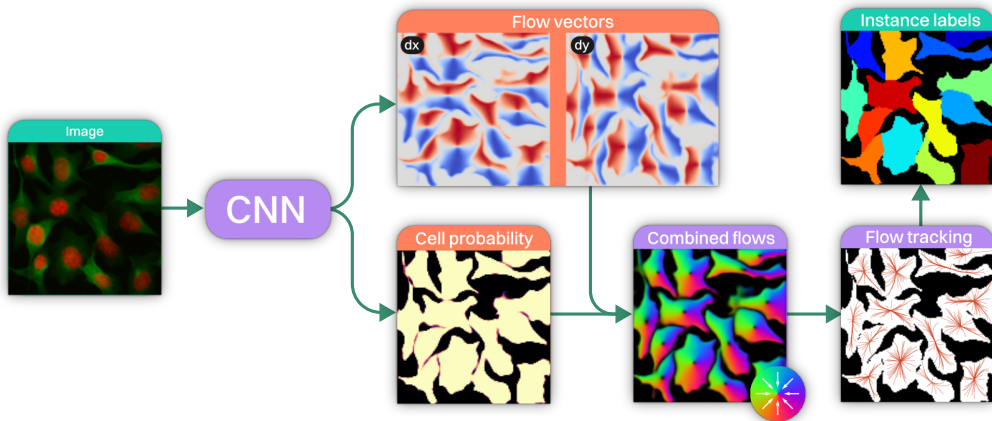


Convolutional Neural Networks: Instance segmentation

- Convolutional Neural Networks learn to extract features from images
- Relevant information is extracted from the features
- Image to image translation / Local by construction



Cellpose: Instance segmentation as flow tracking



Cellpose: Instance segmentation as flow tracking

Cellpose in short

- For **cytoplasm** and **nuclei** instance segmentation
- Trained on microscopy **fluorescence** images
- Limited to **blob-like** & **non-overlapping** cells
- Based on a lightweight **Convolutional Neural Network** $\implies$ easy to fine-tune
True for cellpose 3. Cellpose-SAM is much more computationally demanding.

PART 4

Practical

Practical

On the github go to
[ECI/Day2_Cellpose/Day2_Cellpose.md](#)



Cellpose for machine learning in image analysis

Instructor: Julie Mabon, Post-doc, Unité d'Analyse d'Images Biologiques, Institut Pasteur

Note

In this course we use Cellpose 3.0 instead of the latest Cellpose-SAM as it is faster to run on any laptop and easier to re-train.



Getting ready

Warning

Be sure to have all tools installed as instructed in [Installation](#)



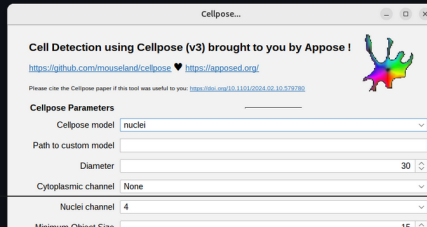
Part 1: Cellpose-Apouse on Fiji

Use case: Segment nuclei in the image, and count spot intensity in each nucleus, then export a CSV using the [Cellpose - Apouse Fiji plugin](#).

1. Load the image
2. Launch the Cellpose plugin:
 - Plugins > Segmentation > Cellpose-Apouse > Cellpose...
3. Set up parameters as suggested. You can hover your mouse over parameters to see what they do.

Important

- Choose appropriate `Cellpose Model`: here we use `nuclei` since we are segmenting nuclei; for cytoplasm use `cyto3`.
- Set Cytoplasmic/Nuclei channel according to your image: here we don't have a cytoplasmic channel (→ None), and the nuclei staining is in channel 4
- Make sure to check `return ROIs` to get ROIs in Fiji for our measurements



Cellpose: Instance segmentation as flow tracking

